FactoryOS
POWERED BY INTELLIGENCE

PRODUCT BROCHURE
NOVEMBER 2026

INTELLIGENT MANUFACTURING PLATFORM

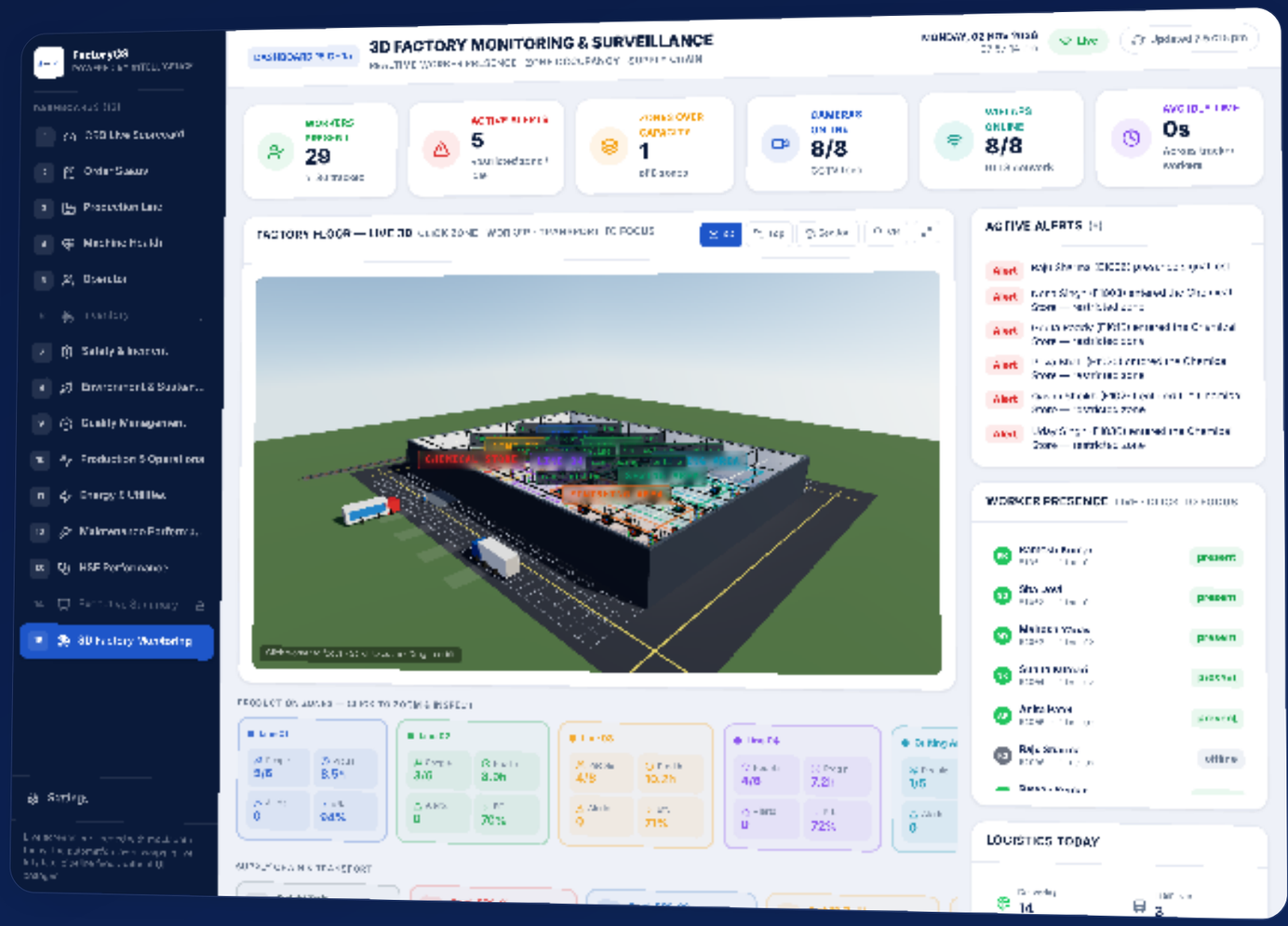
Every signal from your factory floor, in one place.

A real-time operations command center — production, machine health, workforce, safety, quality, energy, and a live 3D digital twin of the plant floor — unified under one configurable, white-label platform.

13
LIVE DASHBOARDS

11
OPERATIONAL MODULES

1
LIVE 3D DIGITAL TWIN



FactoryOS — Intelligent Manufacturing Platform
Prepared as a client-facing capability overview.

PRODUCTION · RELIABILITY · SAFETY · SUSTAINABILITY · SPATIAL INTELLIGENCE

INTRODUCTION

What FactoryOS is

A single platform that replaces the spreadsheets, point tools, and tribal knowledge a plant's leadership, production, maintenance, safety, and quality teams would otherwise maintain separately.

FactoryOS is a real-time factory operations command center built for discrete and process manufacturing plants. It consolidates the metrics a plant would normally track across disconnected tools — production throughput, machine condition, operator performance, order fulfillment, safety incidents, environmental compliance, and energy consumption — into thirteen purpose-built live dashboards, plus a navigable 3D digital twin of the physical factory floor.

Every screen refreshes on its own live cycle, so a number on the page is never more than seconds stale. The platform is architected data-source-agnostic: today's build runs on a realistic simulation layer, with every data type already shaped so a production ERP, MES, SCADA system, or IoT/computer-vision pipeline can be connected later without redesigning a single screen.

Brand identity — name, tagline, logo, and accent color — is runtime-configurable from within the product itself, making FactoryOS deployable as a genuine multi-tenant SaaS offering: the same codebase serves any manufacturing customer under its own identity.

WHO IT SERVES

Plant leadership — one roll-up screen for revenue, efficiency, quality, safety, and energy.

Production & operations managers — line-level throughput, OEE, and downtime.

Maintenance & reliability engineers — condition-based asset health and work-order compliance.

EHS teams — incident, near-miss, and environmental-limit tracking.

Security & floor operations — live spatial awareness of every tracked worker and asset.

BUILT ON

React 18 + TypeScript, Vite, Tailwind CSS, Recharts for 2D visualization, and three.js / React Three Fiber for the live 3D factory model — a modern, maintainable stack chosen for both performance and long-term extensibility.

PLATFORM OVERVIEW

Eleven modules, one navigation

Every module lives behind a single, numbered sidebar — the same navigation shape scales from today's 13 live screens to the two already-reserved slots below.

01
Executive Intelligence
CEO Live Scorecard

02
Order Management
Order Status

03
Production
Production Line · Production & Operations

04
Asset & Machine Mgmt
Machine Health · Maintenance Performance

05
Workforce
Operator

06
Safety & Compliance
Safety & Incident · HSE Performance

07
Sustainability
Environment & Sustainability

08
Quality
Quality Management

09
Utilities
Energy & Utilities

10
Spatial Intelligence
3D Factory Monitoring & Surveillance

11
Administration
Settings — Brand & Theme

—
Inventory — Planned
Spares & Stock

DASHBOARD	ROUTE	STATUS	PRIMARY TABLE
CEO Live Scorecard	/d/overview	LIVE	—
Order Status	/d/orders	LIVE	15-column order tracking table
Production Line	/d/production-line	LIVE	Plant layout performance map
Machine Health	/d/machine-health	LIVE	12-column machine list (980 assets)
Operator	/d/operator	LIVE	12-column operator roster (1,248 people)
Inventory	— (locked)	COMING SOON	—
Safety & Incident	/d/safety	LIVE	9-column incident log
Environment & Sustainability	/d/environment	LIVE	7-column live parameter table
Quality Management	/d/quality	LIVE	11-column line/area quality table
Production & Operations	/d/production-ops	LIVE	9-column unit/shop production table
Energy & Utilities	/d/energy-utilities	LIVE	14-column utility consumption table
Maintenance Performance	/d/maintenance	LIVE	15-column area & shutdown tables
HSE Performance	/d/hse	LIVE	14-column plant & risk-control tables
Executive Summary	— (locked)	COMING SOON	—
3D Factory Monitoring & Surveillance	/d/factory-3d	LIVE	Live worker presence list

PART I

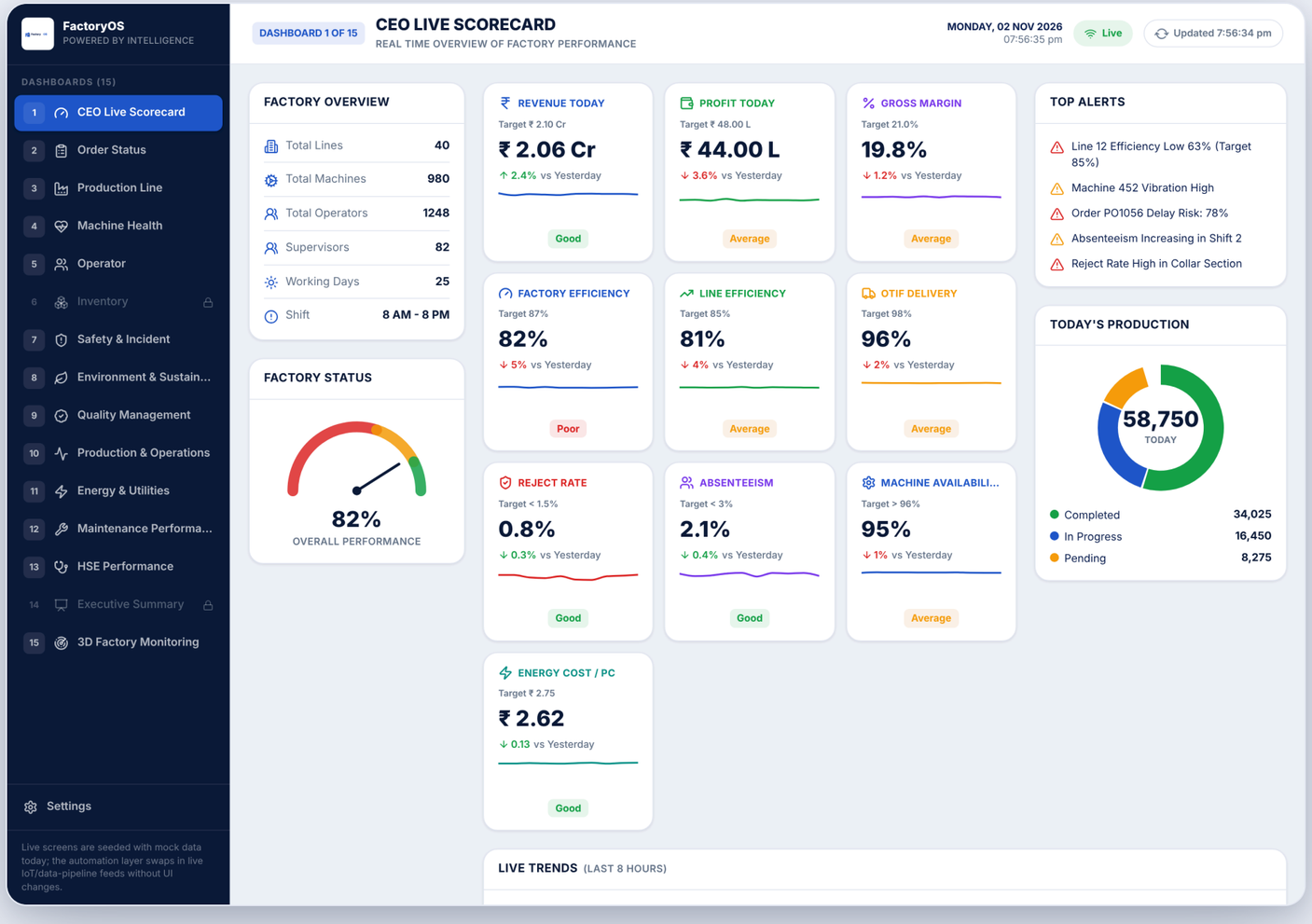
Executive & Commercial Intelligence

The two screens a plant's leadership and commercial team open first: a factory-wide performance roll-up, and a live view of every purchase order from PO to dispatch.

01 — Executive Intelligence
02 — Order Management
03 — Production
04 — Asset & Machine Mgmt
05 — Workforce
06 — Safety & Compliance
07 — Sustainability
08 — Quality
09 — Utilities
10 — Spatial Intelligence
11 — Administration

CEO Live Scorecard

A single roll-up screen answering "are we making money, are we efficient, and what needs my attention today" — without opening a spreadsheet.



Revenue Today ₹2.06 Cr **Factory Efficiency 82%** **OTIF Delivery 96%** **Reject Rate 0.8%** **Machine Availability 95%**

USER WORKFLOW

1. Scan status tags for red/amber across the ten KPI cards.
2. Read the Top Alerts feed for the specific line, machine, or order flagged.
3. Drill into the relevant dedicated dashboard to act.

BUSINESS VALUE

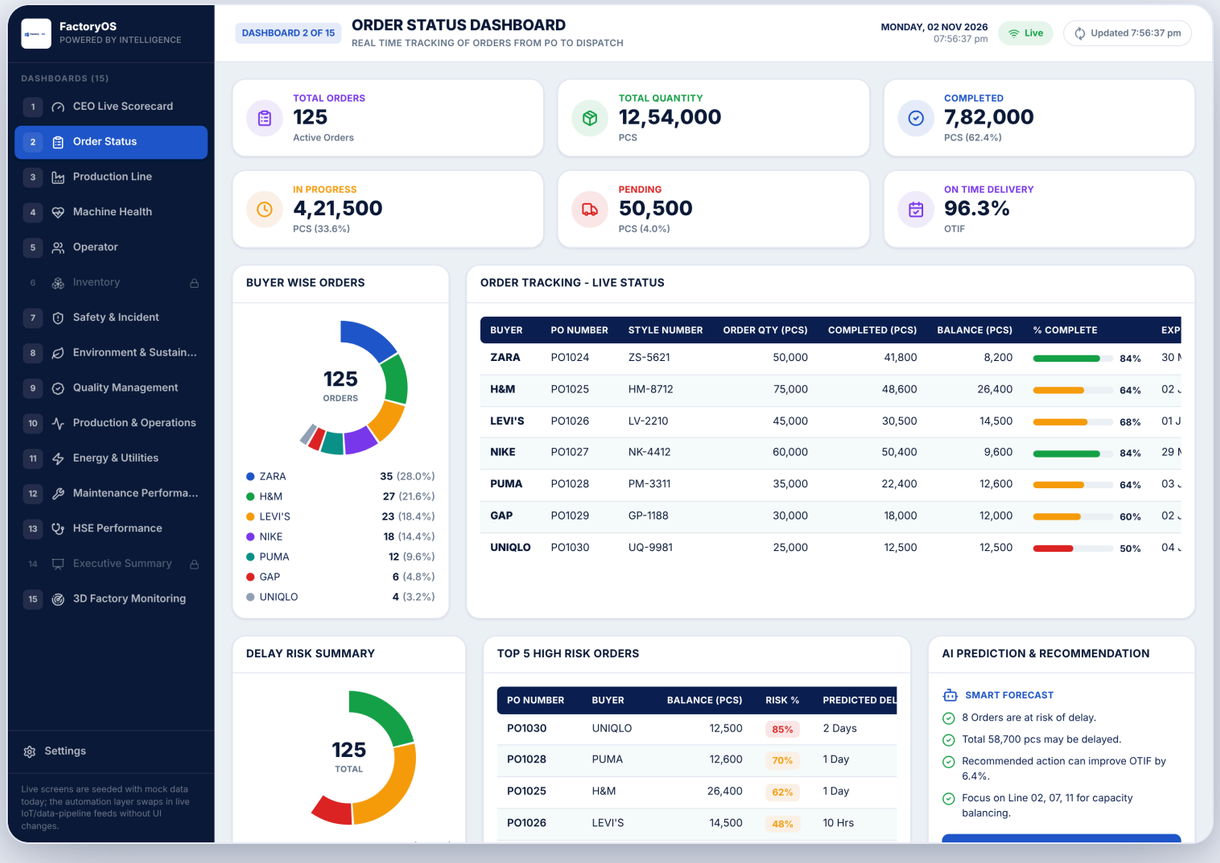
Collapses revenue, efficiency, quality, safety, and energy — normally owned by five different teams — into one screen, so leadership spends minutes, not meetings, finding out what's wrong today.

DESIGN NOTES

Ten KPI cards each carry a trend sparkline and a vs-yesterday delta, not just a static number — every metric shows direction, not only magnitude.

Order Status

Tracks every purchase order from PO issuance to dispatch, with AI-assisted delay-risk prediction and a suggested action on every row.



01 · Risk-Ranked Order Table

15-column live order tracking

Buyer, PO/style number, quantity complete vs. balance, expected dispatch, risk %, delay prediction, current line/operator, and a suggested action — per order.

02 · Delay Intelligence

Delay Risk Summary & Top 5 High-Risk Orders

Surfaces which open POs are most likely to miss dispatch before the delay happens, ranked by risk percentage.

03 · AI Recommendation Panel

Prediction with a "View Detailed AI Insights" action

Pairs every flagged order with a concrete next step, not just a warning.

Business Value

Act before the customer is affected

A planner reassigns an at-risk order to a different line or operator while there's still time to recover, instead of discovering the delay at dispatch.

PART II

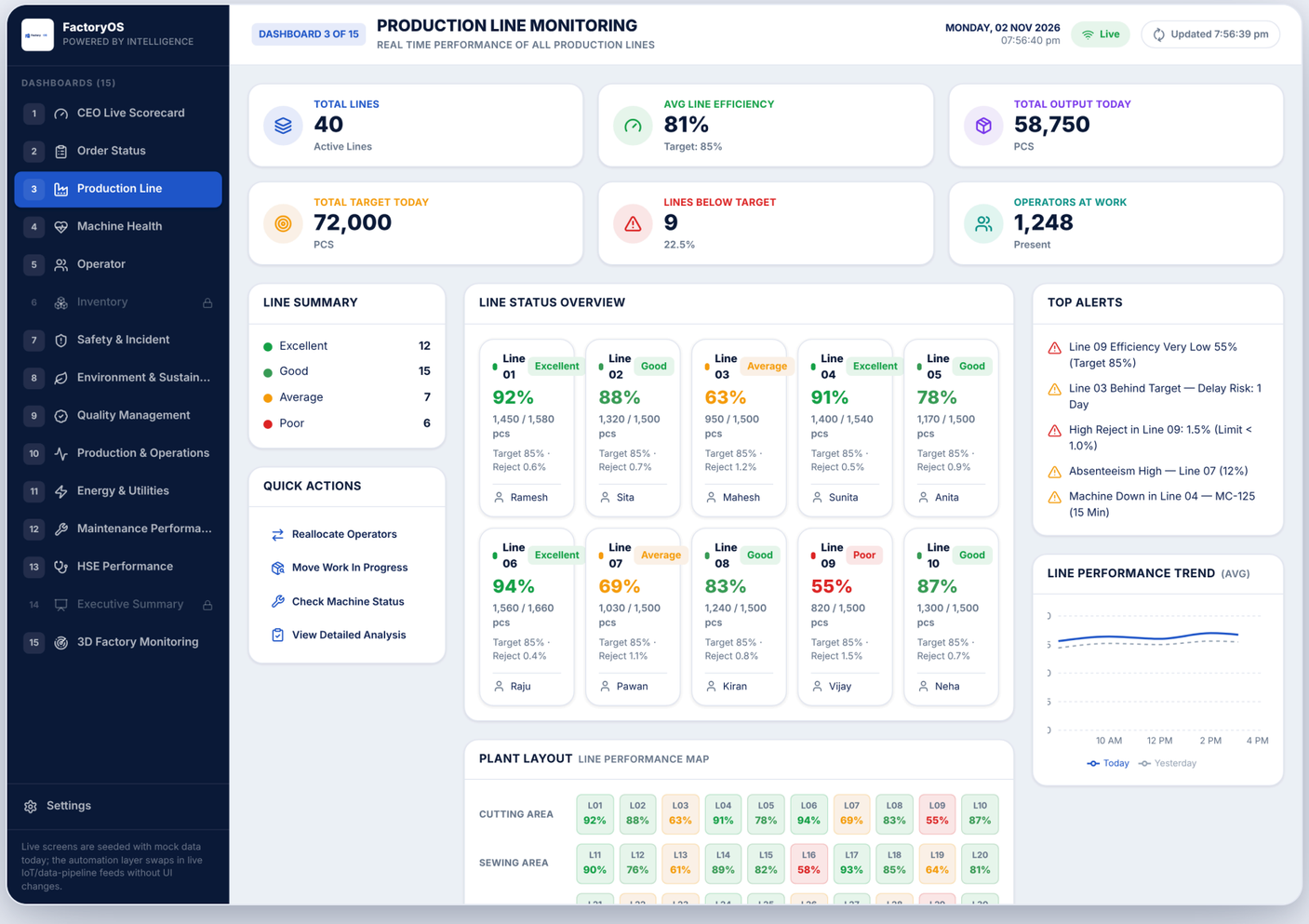
Production & Asset Operations

Line-level throughput and OEE, the 980-machine fleet's condition and remaining useful life, and the work-order discipline that keeps both running.

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Production Line

Real-time throughput against target for every production line, with a spatial plant-layout heat map.



Total Lines 40 **Avg Efficiency 81%** **Lines Below Target 9** **Operators at Work 1,248**

USER WORKFLOW

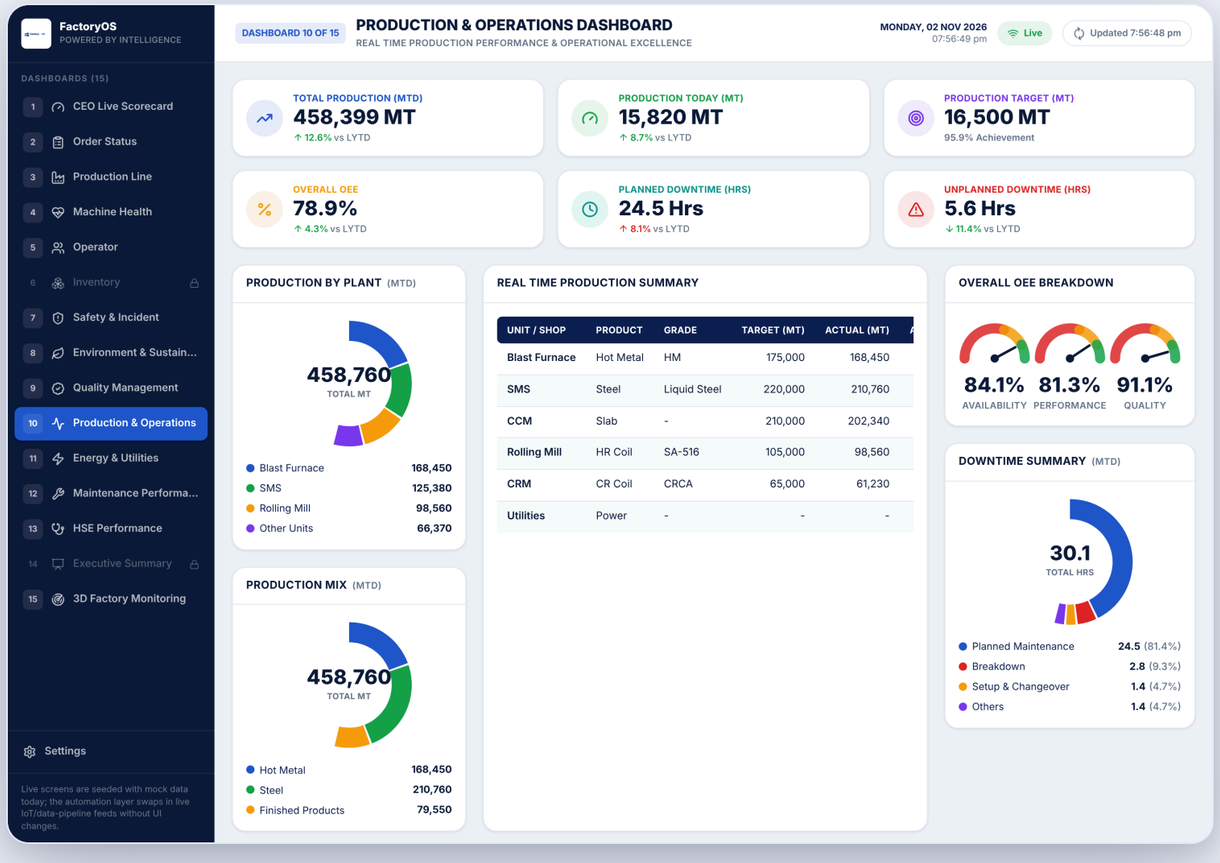
1. Scan the Plant Layout heat map for underperforming cells.
2. Cross-reference the Top Alerts feed for the affected line.
3. Trigger a Quick Action (e.g. reallocate operators) without leaving the screen.

BUSINESS VALUE

Turns 40 individual lines into one glanceable spatial map, so a shortfall is visible immediately rather than discovered at end-of-shift reporting.

Production & Operations

The operational counterpart to the Scorecard — unit-level production, OEE, and a downtime split that isolates where output is actually being lost. The platform's most data-dense screen, with fourteen coordinated panels.



01 · Unit-Level Production Table

9-column Real Time Production Summary

Unit/shop, product, grade, target vs. actual MT, achievement %, OEE %, and status, benchmarked against yesterday.

02 · Downtime Split

Planned vs. unplanned downtime, isolated

Separates the two at the unit level — the single most common blind spot in plant reporting, since OEE is lost almost entirely in the unplanned bucket.

03 · OEE Everywhere

OEE Breakdown, OEE Trend, and OEE by Unit

The same composite metric tracked three ways — as a snapshot, a 7-day trend, and a per-unit comparison.

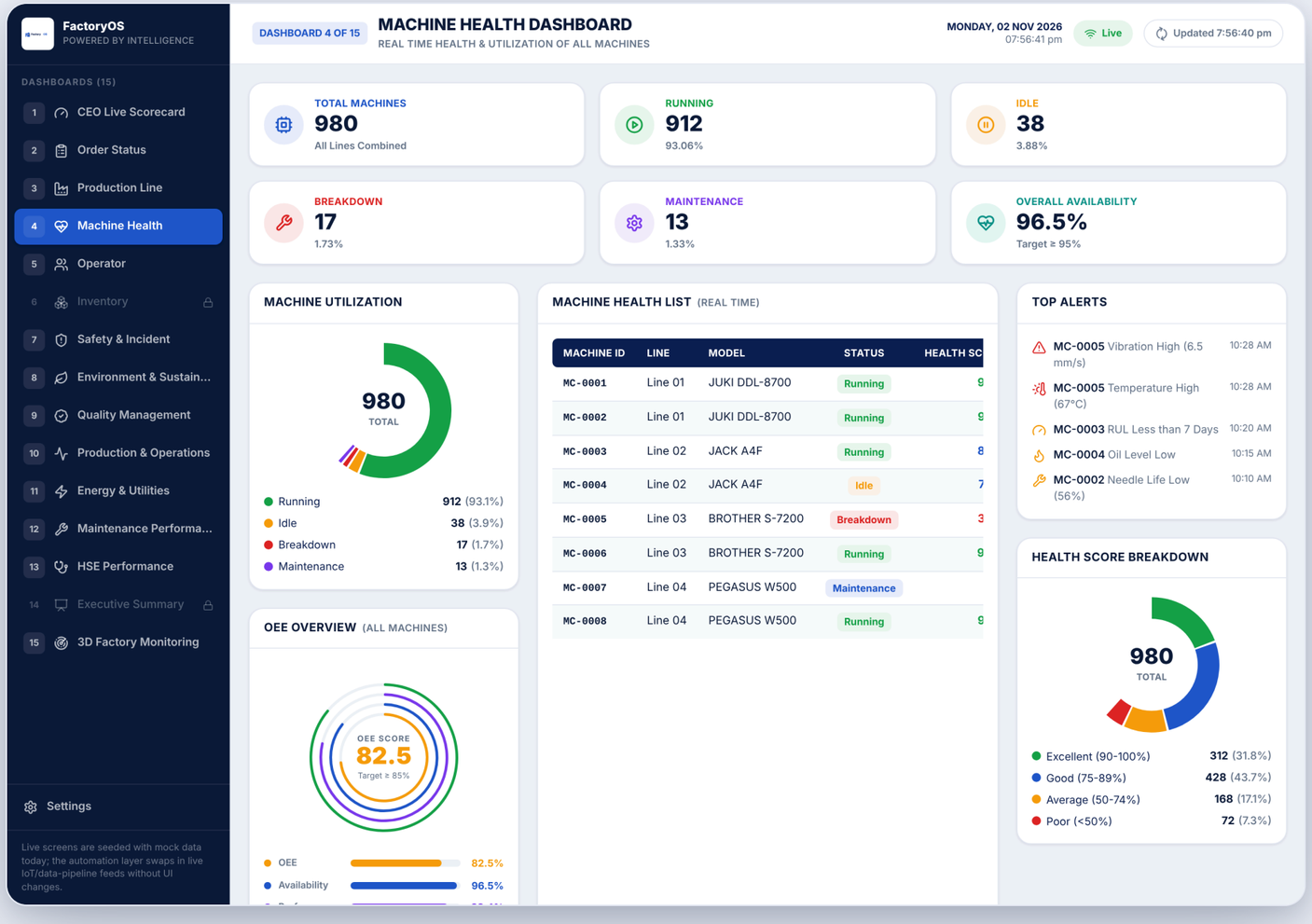
Business Value

Find the underperforming unit in seconds

An operations director compares Target Achievement and OEE by Unit, then the Top 5 Downtime Reasons, to decide whether the gap is maintenance load or unplanned failure.

Machine Health

Condition-monitoring and OEE for the full 980-machine fleet, centered on a concentric radial OEE gauge that refreshes live.



980 Machines **912 Running** **Availability 96.5%**

USER WORKFLOW

1. Check the OEE gauge for the fleet-wide composite score.
2. Filter the 12-column machine list for Breakdown status or RUL under 7 days.
3. Schedule intervention before a failure stops the line.

BUSINESS VALUE

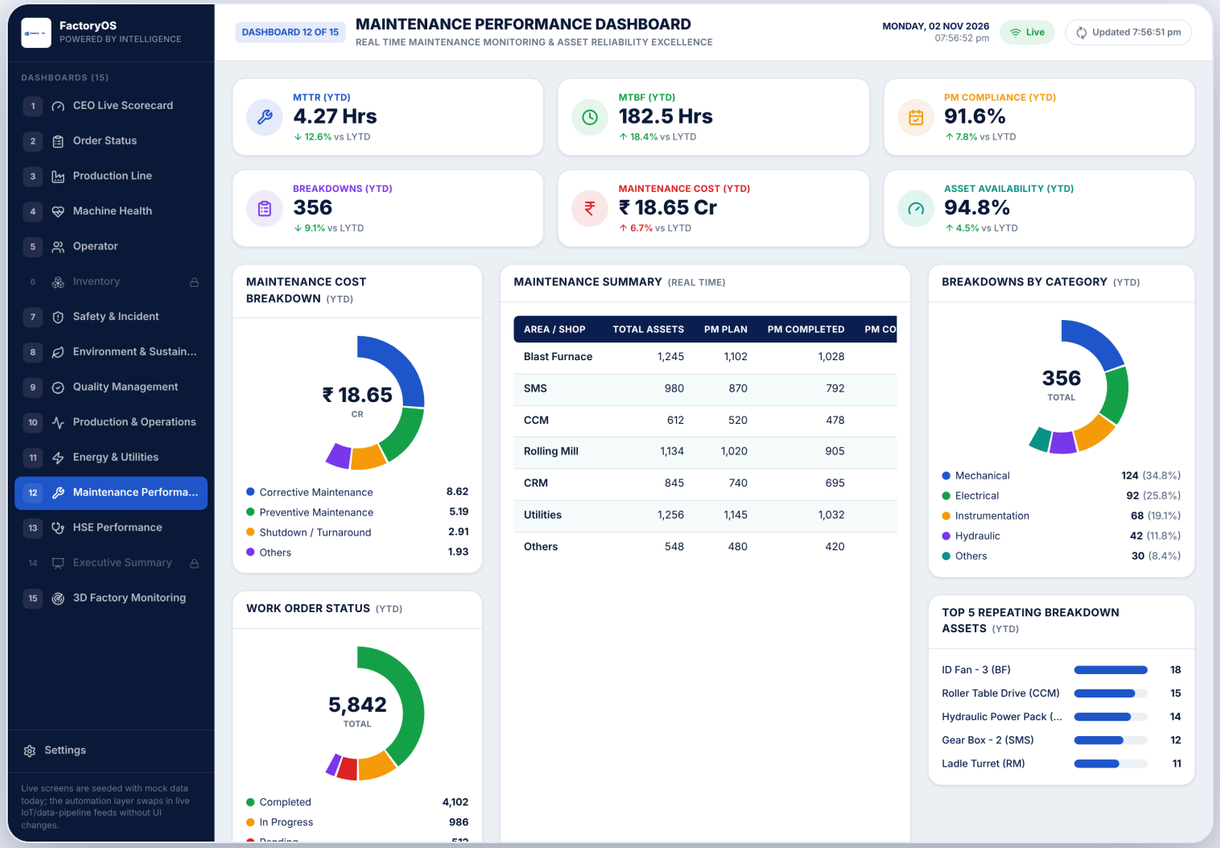
Converts raw readings — vibration, temperature, running hours — into an actionable health score and remaining-life estimate per machine, so maintenance is scheduled on condition, not a fixed calendar.

DESIGN NOTES

The OEE Overview reads Availability × Performance × Quality = OEE as one concentric gauge with an inline legend — the composite metric and its three drivers in a single glance.

Maintenance Performance

Work-order and reliability tracking — MTTR, MTBF, PM compliance, and cost — at the area/shop level.



01 · Reliability Table

15-column area-level Maintenance Summary

Total assets, PM plan vs. completed, compliance %, breakdowns, MTTR, MTBF, and cost per area.

02 · Repeat-Failure Detection

Top 5 Repeating Breakdown Assets

The highest-leverage maintenance signal — a recurring failure on the same asset is a root-cause problem, not a random event.

03 · Shutdown Planning

Planned Shutdowns / Turnaround schedule

Area, start/end date, and progress for every scheduled outage.

Business Value

Shift from reactive to preventive

Cross-referencing repeat-failure assets against PM compliance and the Scheduled-vs-Unscheduled split justifies the investment case for more preventive maintenance.

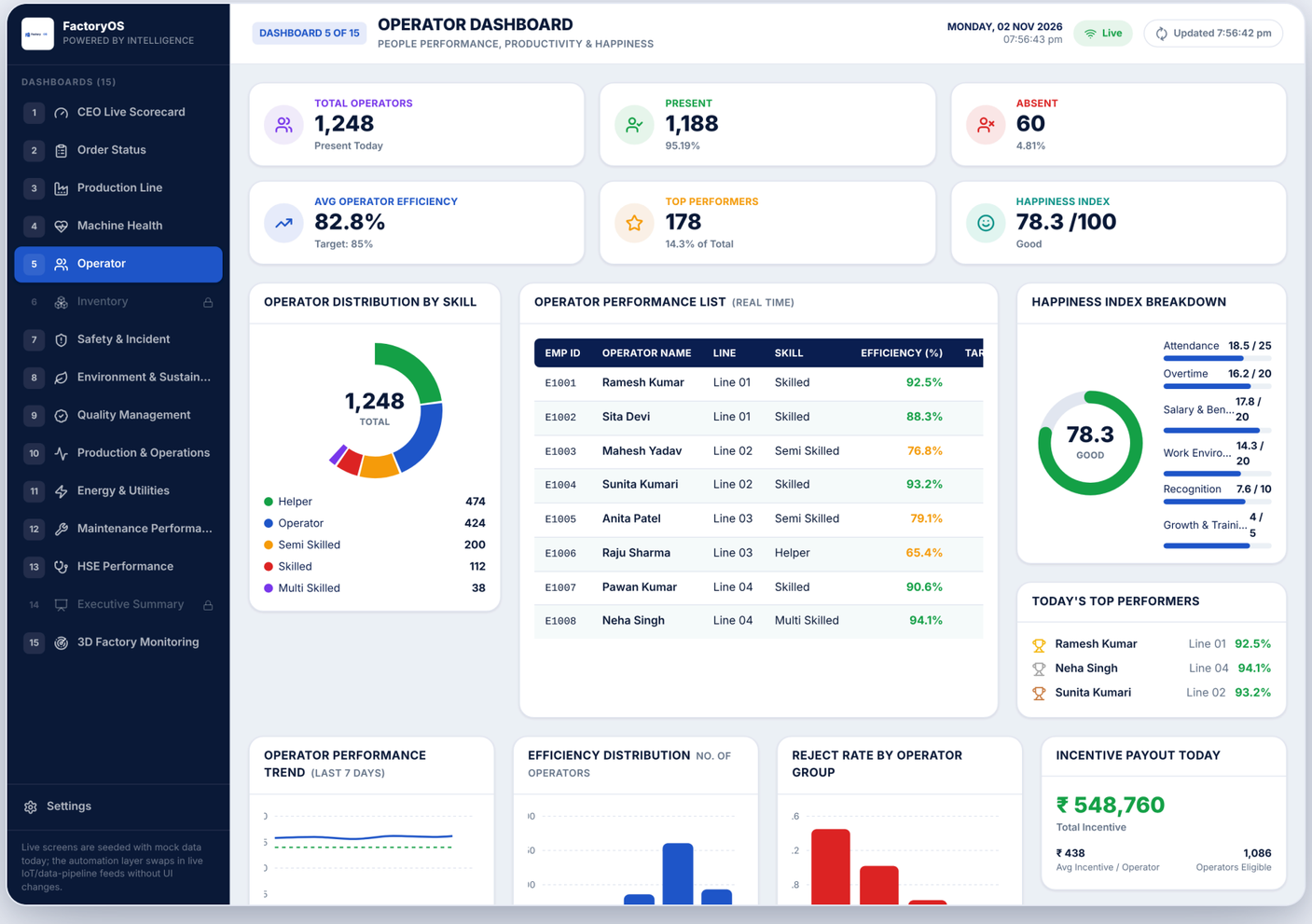
PART III

Workforce, Safety & Compliance

1,248 operators' performance and wellbeing, and the leading and lagging indicators that keep every one of them safe.

Operator

People-performance and attendance tracking across the 1,248-person operator workforce, linking productivity to engagement.



1,248 Operators **1,188 Present** **Happiness 78.3/100**

USER WORKFLOW

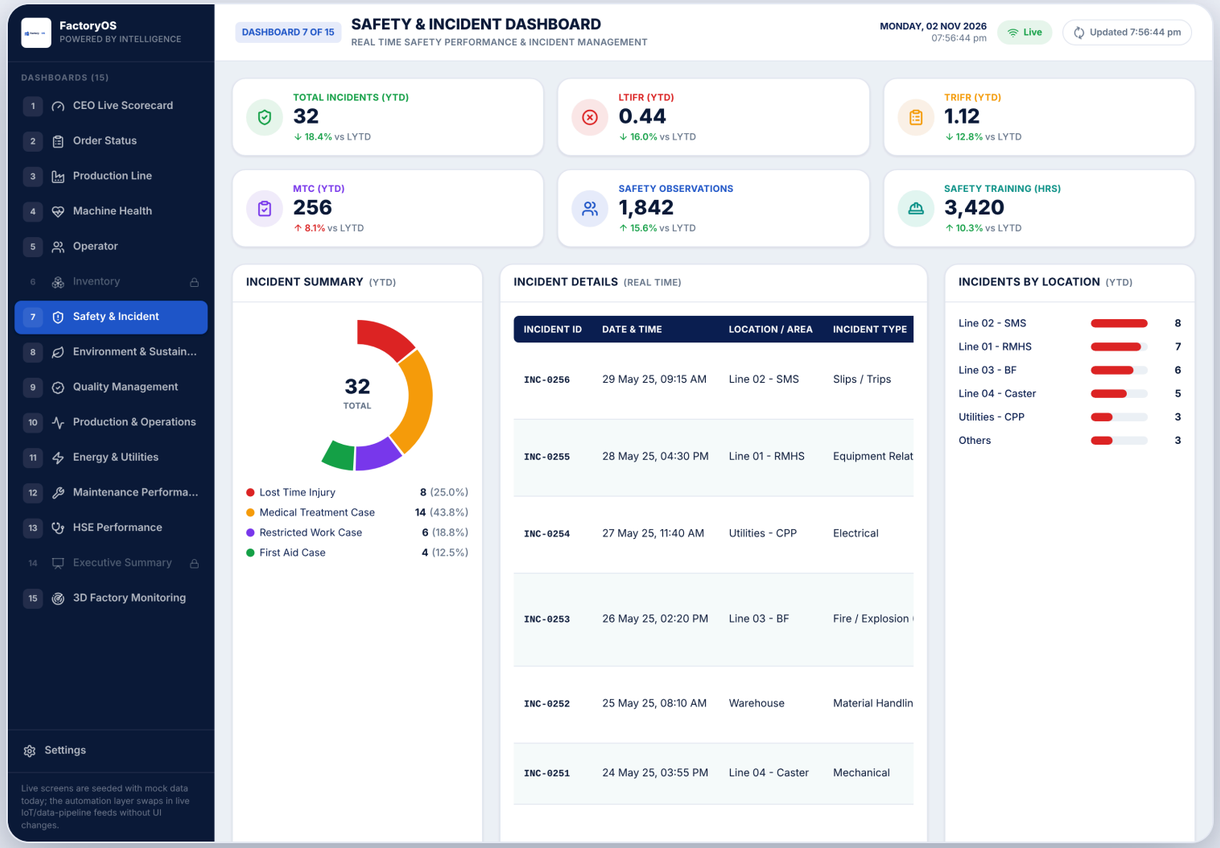
1. Review the performance list for operators below efficiency target.
2. Check happiness score alongside output and reject rate.
3. Use Top Performer and incentive data to plan recognition or coaching.

BUSINESS VALUE

Links performance to engagement and pay in one roster, making the connection between workforce wellbeing and productivity visible, not anecdotal.

Safety & Incident

Real-time incident logging and lagging-indicator tracking (LTIFR / TRIFR) against a target trend line.



01 · Incident Log

9-column live incident table

Date/time, location, type, severity, description, LTI potential, status, and days open per incident.

02 · Leading Indicators

Safety Observations & Near Miss Reporting

Tracks the proactive signals — observations and near-misses reported — alongside actual incidents.

03 · Trend vs. Target

12-month LTIFR trend with target line

Confirms whether the plant is improving month over month against a defined goal, not just a raw count.

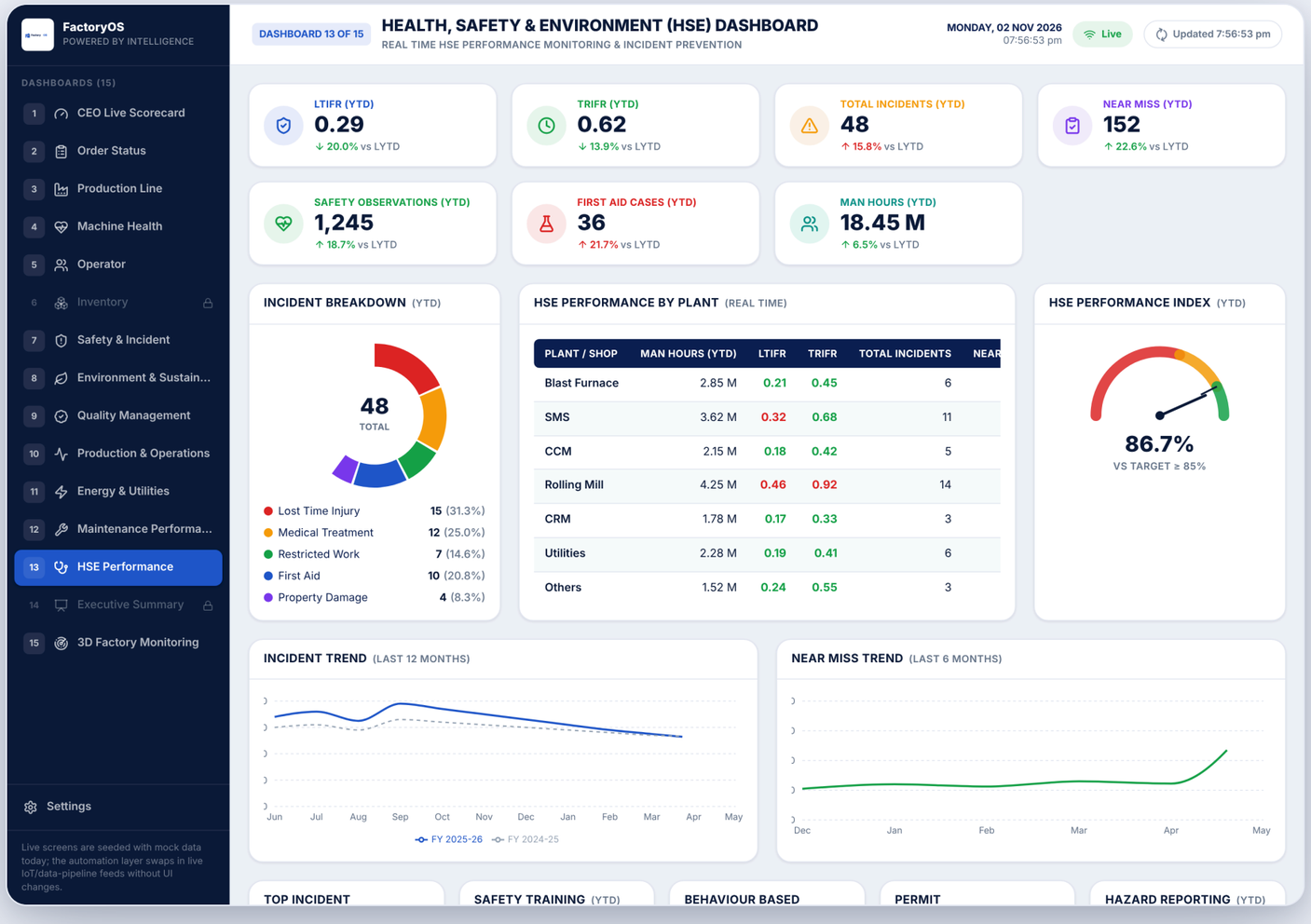
Business Value

Proactive, not reactive, safety

Makes the gap between leading and lagging indicators visible on one screen — the foundation of a mature safety program.

HSE Performance

Plant-wide Health, Safety & Environment compliance — including behaviour-based safety and critical-control effectiveness.



LTIFR 0.29 **Near Miss 152** **18.45M Man Hours**

USER WORKFLOW

1. Review Critical Risk Control Status for any control marked ineffective.
2. Prioritize closing that gap first.
3. Track the plant-by-plant table to see which site needs support.

BUSINESS VALUE

The Critical Risk Control effectiveness breakdown is a maturity signal most safety dashboards omit — it answers whether controls are actually working, not just how many incidents occurred.

PART IV

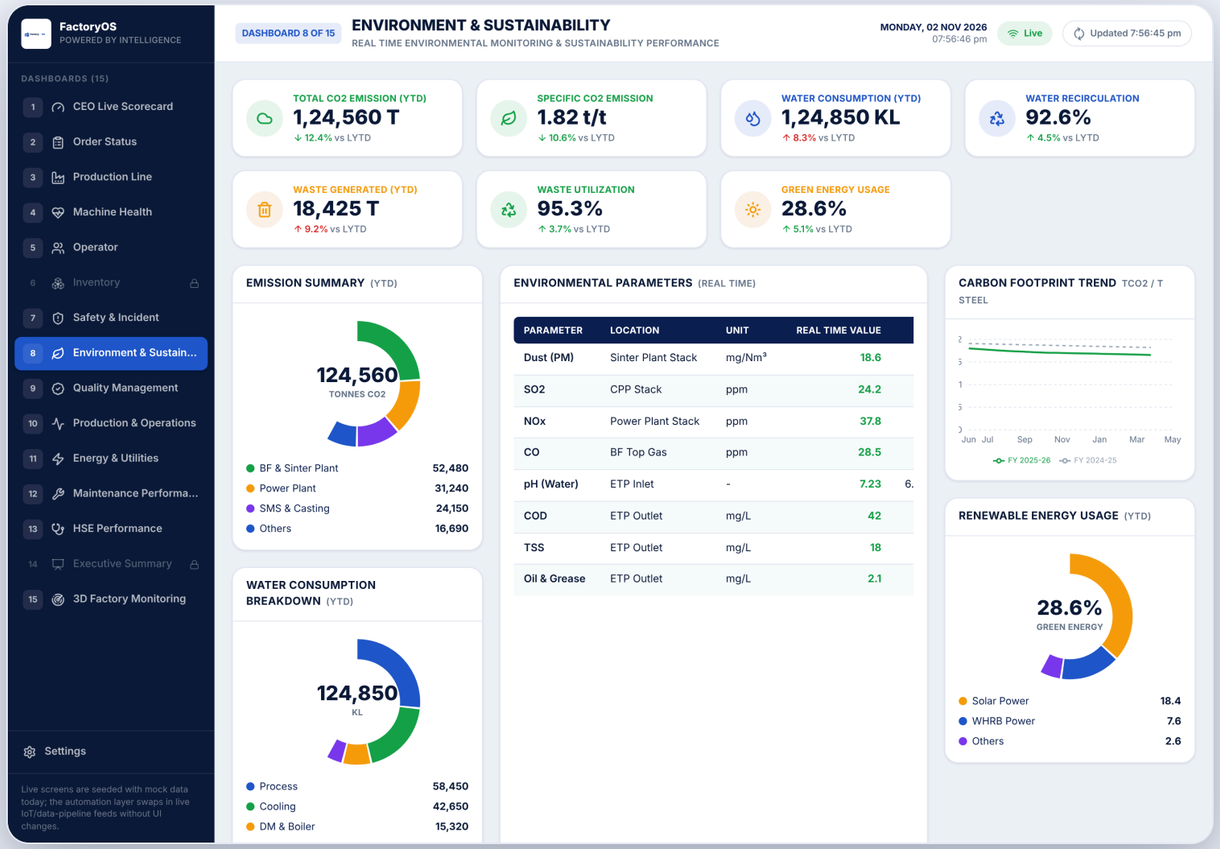
Sustainability, Quality & Utilities

Environmental compliance against regulatory limits, statistical process control against customer expectations, and the resource cost of running the plant.

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Environment & Sustainability

Environmental compliance and sustainability metrics — emissions, water, waste, and renewable energy mix.



01 · Live Compliance Table

7-column Environmental Parameters

Each regulatory parameter's real-time value tracked directly against its limit, with a clear status flag.

02 · Resource Intensity

Specific CO_2 emission & water recirculation %

Normalized per-tonne metrics, not just absolute totals — the figures that actually drive ESG reporting.

03 · 12-Month Trends

CO_2, water, and waste trend charts

Long-term trajectory tracking alongside the day-to-day compliance table.

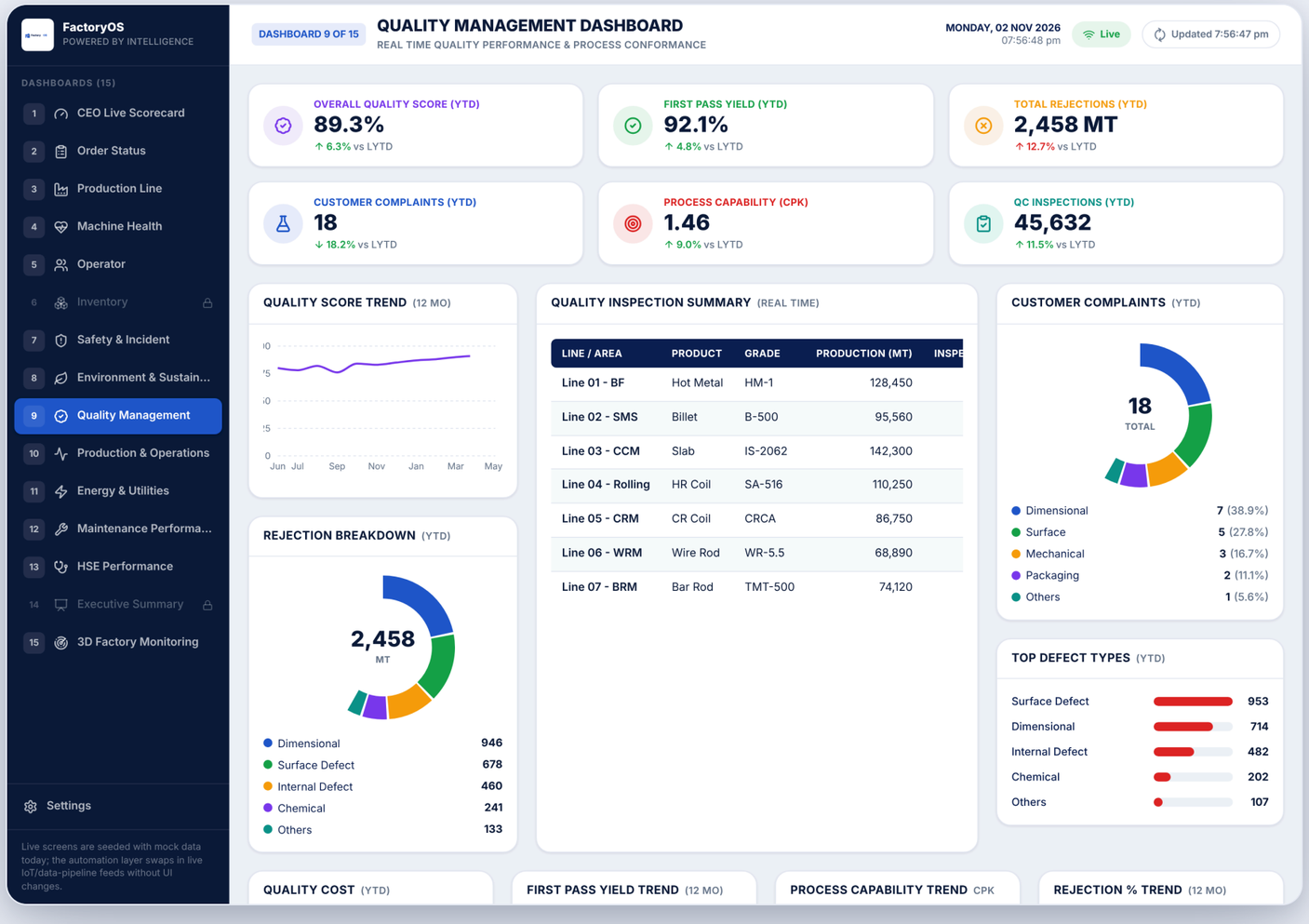
Business Value

One source for daily and annual reporting

Real-time regulatory compliance and long-term sustainability trend reporting share one data source, not two.

Quality Management

Plant-wide conformance — first pass yield, rejection rate, process capability, and customer complaints.



Quality Score 89.3% **FPY 92.1%** **Cpk 1.46**

USER WORKFLOW

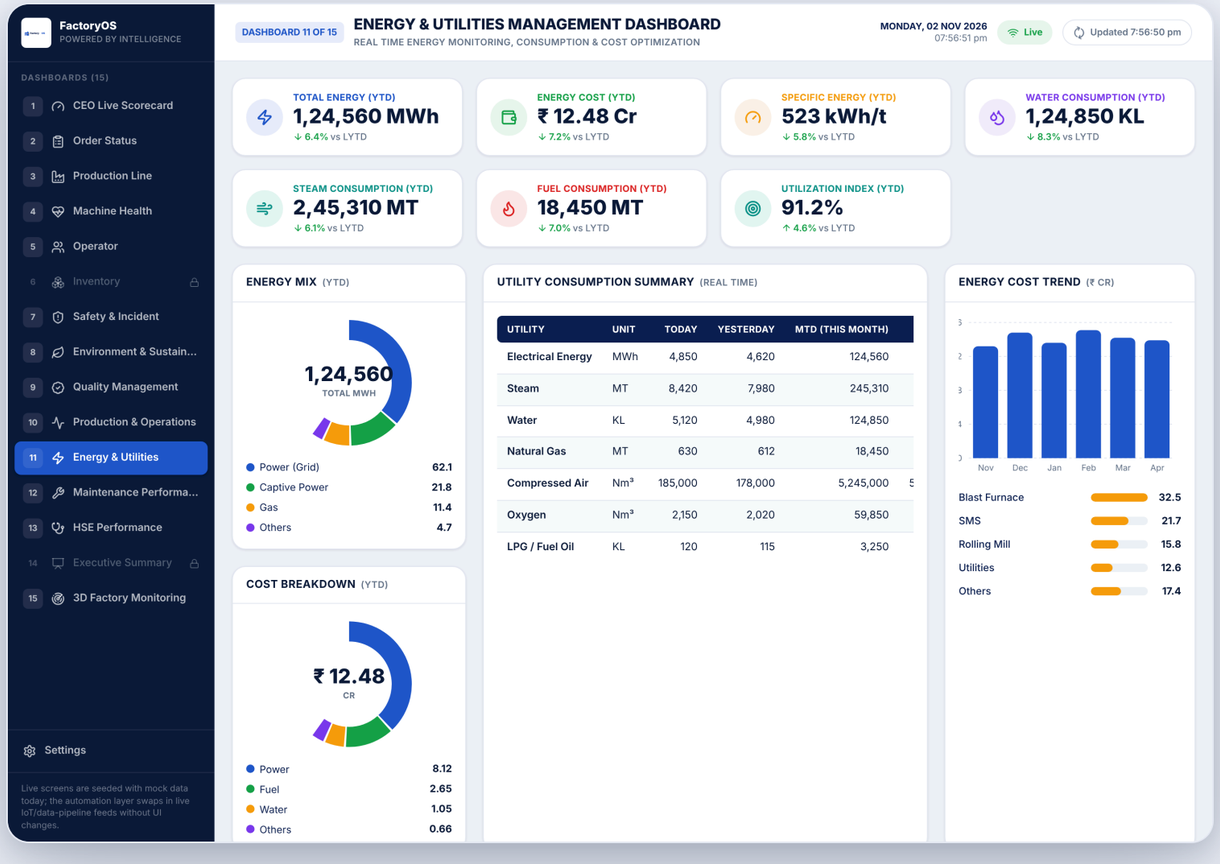
1. Scan the line/area table for Cpk below target or a rejection spike.
2. Cross-reference the Top Defect Types ranking.
3. Identify the recurring failure mode to address.

BUSINESS VALUE

Connects statistical process control directly to commercial outcomes, so a capability problem is caught as a quality-engineering signal before it becomes a customer complaint.

Energy & Utilities

Consumption, cost, and efficiency across energy, water, steam, and fuel — with identified savings opportunities.



01 · Utility Table

14-column consumption summary

Today, yesterday, MTD, LY MTD, variance, YTD, and LYTD for every tracked utility.

02 · Savings Opportunities

Opportunity paired with potential ₹/year savings

Every identified opportunity carries its own MWh and ₹ Cr-per-year savings estimate.

03 · Plant Benchmarking

Plant Wise Specific Energy ranking

kWh per tonne of steel, ranked across plants to find the outlier.

Business Value

From cost reporting to a prioritized action list

A rupees-per-year figure next to every opportunity turns energy management into a ranked investment decision.

PART V

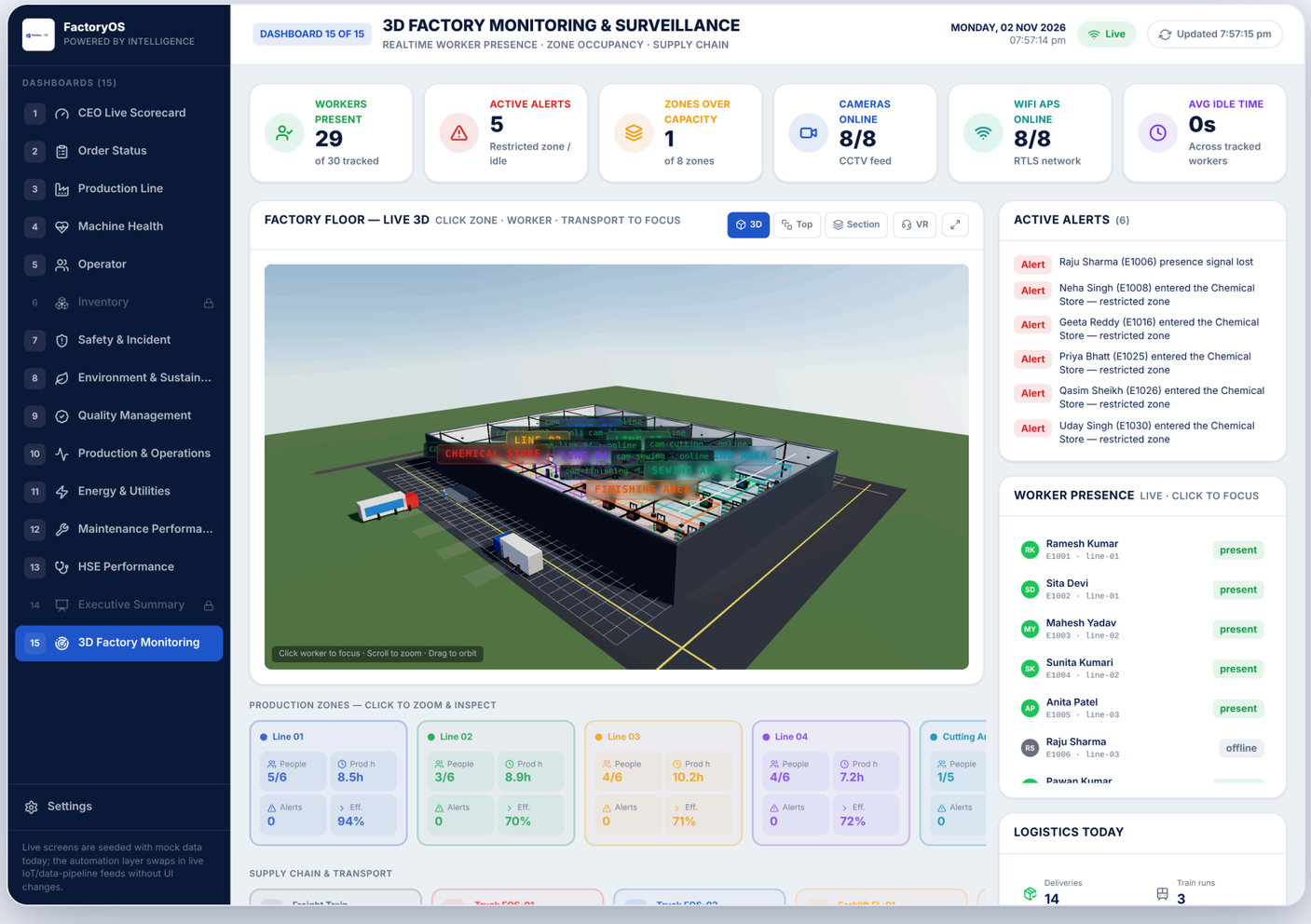
Spatial Intelligence & Administration

The platform's signature capability — a live, navigable 3D digital twin of the factory floor — and the branding controls that make FactoryOS a true multi-tenant product.

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3D Factory Monitoring & Surveillance

A live, navigable 3D digital twin of the factory floor — exactly where every tracked worker, CCTV camera, WiFi access point, and transport asset is right now.



01 · Four View Modes

Perspective · Top-Down · Cross-Section · VR

Orbit-controllable 3D model with a cutaway view and WebXR immersive mode, hardware permitting.

02 · Click-to-Focus Presence

Worker list → camera flies to location

Click any name in the live Worker Presence list and the 3D camera flies to their exact position in the model.

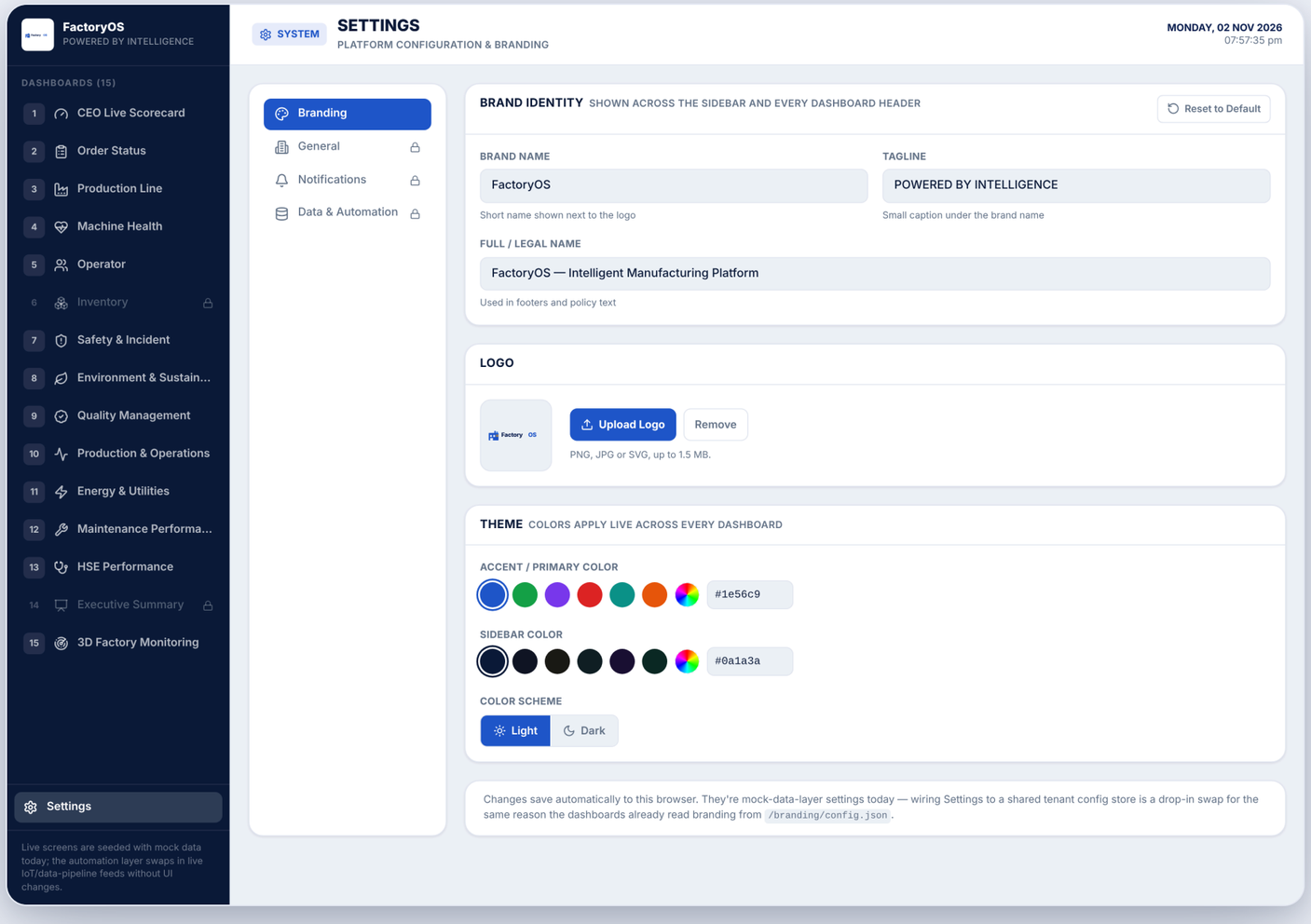
03 · Zone & Logistics Intelligence

Restricted-zone alerts, per-zone production cards, live logistics

Headcount, productivity, and efficiency per zone, plus deliveries, train runs, truck trips, and forklift activity today.

Settings

White-label branding and theme administration — the mechanism that lets one codebase serve any manufacturing customer under its own identity.



USER WORKFLOW

1. Open Settings when onboarding a new customer.
2. Replace the brand name, tagline, logo, and accent color.
3. Every one of the 13 live dashboards re-renders under the new identity instantly — no reload required.

BUSINESS VALUE

This is what makes FactoryOS a genuine multi-tenant SaaS product rather than a single-customer build — rebranding is a two-minute in-app task, not an engineering request.

ALSO INCLUDED

Live light/dark theme toggle, with every status color intentionally holding its hue across both modes so charts read identically either way.

USER EXPERIENCE & DESIGN

One visual system, every screen

A single centralized token file drives color, radius, shadow, and type across all thirteen dashboards and the 3D layer — nothing is restyled per screen.

BRAND & STATUS PALETTE

#1E56C9	#0A1A3A	#16A34A	#F59E0B	#DC2626	#7C3AED	#0D9488	#2563EB
BRAND BLUE	NAVY 950	STATUS GOOD	STATUS WARN	STATUS BAD	STATUS PURPLE	STATUS TEAL	STATUS INFO

Every status color keeps the same hue in light and dark mode — only backgrounds flip — so a chart reads identically regardless of theme.

TYPOGRAPHY — INTER

82%

FACTORY EFFICIENCY

Body copy sits at 12–12.5px for density; card titles are bold, uppercase, and letter-spaced to separate structure from data at a glance — tuned for a control-room monitor, not a marketing page.

DATA TABLE — ZEBRA STRIPING, ALIGNED COLUMNS, INLINE STATUS

MACHINE ID	STATUS	HEALTH	VIBRATION
MC-0001	Running	96%	2.1
MC-0005	Breakdown	32%	6.5
MC-0007	Maintenance	—	—

CORE COMPONENTS

STATUS BADGE

Running Idle Breakdown Maintenance

KPI CARD

OVERALL AVAILABILITY
96.5%

CARD CONTAINER

1rem radius, 1px hairline border, two-layer soft shadow — the single `.card` surface every panel, table, and chart container in the product uses.

LIGHT / DARK MODE — SAME HUE, DIFFERENT SURFACE

Toggled live from Settings. Status colors are intentionally identical in both modes — only page and card backgrounds flip.

LIGHT

MACHINE HEALTH

Running Breakdown

DARK

MACHINE HEALTH

Running Breakdown

3D LAYER CONSISTENCY

The 3D Factory Monitoring view reuses these same status tokens for worker-avatar and device-marker colors, so the spatial view and the 2D dashboards read as one visual system, not two products bolted together.

PLATFORM WORKFLOWS

From signal to action

Two representative end-to-end workflows the platform supports today, as implemented — not aspirational.

WORKFLOW 1 — RESTRICTED-ZONE RESPONSE (SPATIAL INTELLIGENCE)

1

Alert fires

A worker enters the Chemical Store restricted zone; the 3D Factory Monitoring Active Alerts feed logs it in real time.

2

Operator clicks the name

The flagged worker's name is clicked in the live Worker Presence list.

3

Camera flies to location

The 3D camera animates to the worker's exact position in the model for visual confirmation.

4

Confirm via CCTV

The nearest camera marker's online status is checked to confirm footage is available.

5

Respond

Floor operations dispatches a response with full spatial context already established.

WORKFLOW 2 — NEW CUSTOMER ONBOARDING (ADMINISTRATION)

1

Open Settings

An administrator onboarding a new customer navigates to the Settings page.

2

Set brand identity

Brand name, tagline, and full/legal name are entered.

3

Upload logo

The customer's logo replaces the default mark.

4

Choose theme

Accent color and light/dark mode are set to match the customer's brand.

5

Live everywhere

All 13 dashboards re-render under the new identity instantly — no redeploy.

FactoryOS
POWERED BY INTELLIGENCE

Eleven modules. Thirteen live dashboards. One digital twin of your factory floor.

FactoryOS unifies production, reliability, workforce, safety, quality, sustainability, energy, and spatial intelligence under a single, white-label-ready platform — built to extend to a real data pipeline without a UI rewrite.

DOCUMENT PREPARED NOVEMBER 2026

CLIENT-FACING CAPABILITY OVERVIEW